

FedMEvo: A Joint Agentic Memory Evolution and Communication Optimization Framework for Wireless Federated Agent Collaboration

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Abstract—In the LLM era, federated agent collaboration enables distributed agents to exchange valuable knowledge and improve local task performance. We extend Agentic Memory Evolution (AME) to the federated collaboration framework, giving rise to the federated AME paradigm. However, this direct extension remains insufficiently adapted, with limitations including: (1) memory exchange lacks explicit adaptation to practical wireless resource constraints; (2) the applicability of local memory to other agents’ tasks is insufficiently considered; and (3) the semantic quality of exchanged memories is insufficiently considered. To address these issues, we propose a joint agentic memory evolution and communication optimization framework, namely FedMEvo, for communication-resource-constrained wireless scenarios. Specifically, we first formulate federated AME under explicit wireless communication constraints. We then develop a task-aware co-optimization mechanism for on-demand agentic-memory exchange and wireless resource allocation. Moreover, we propose a multi-granularity memory representation strategy for the co-optimization, enabling joint consideration of semantic information and communication payload across varying representation sizes. Importantly, we provide a rigorous theoretical characterization of the proposed co-optimization, establishing its solution structure and convergence properties. Extensive experiments show that FedMEvo achieves 40.8% higher average task performance than the strongest external baseline and 82.5% lower communication than the full-memory-upload variant. Further evaluations demonstrate robust performance-communication efficiency across diverse experimental settings. We further validate its practical applicability in a real-world industrial IoT scenario.

Index Terms—Federated agent collaboration, agentic memory evolution, wireless communication, Joint optimization, large language models.

I. INTRODUCTION

With the emergence of large language models (LLMs), LLM-based intelligent agents are increasingly employed for perception, reasoning, and decision-making tasks [1], [2]. However, a single agent remains limited when continuously encountering diverse tasks and local observations. *Federated Agent Collaboration* provides a distributed cooperation paradigm in which multiple agents exchange learned knowledge to improve local task performance [3]–[6] (See Fig. 5). However, such collaboration inevitably incurs communication overhead [7]. In real-world wireless environments with limited shared communication resources, this further calls for exchanging lightweight semantic knowledge to reduce communication burden while retaining collaborative benefits [8].

These semantic knowledge representations are compact and reusable abstractions distilled by agents from their local tasks [3]–[6], rather than model-parameters exchanged in conventional federated learning or federated LLM PEFT, or prompts optimized in federated prompt learning [9]. In existing federated agent collaboration, such semantic knowledge has been instantiated in different forms: FICAL [6] exchanges compact knowledge compendiums to support federated in-context agent learning; FederatedSkill [3] communicates semantic skill diffs for personalized skill evolution across agents; FedAgentKE [4] evolves transferable reasoning abstractions through semantic knowledge distillation, aggregation, and adaptation; and FedWorld [5] exchanges structured abstract transition rules for scope-aware world-model federation. *Agentic memory is also an important form of semantic knowledge* [10], [11], compactly representing task-derived experience to support subsequent reasoning and task execution, while helping reduce transmission overhead and raw local data exposure [12], [13]. These properties make it particularly promising for federated collaboration, yet its role as a federated semantic knowledge carrier remains comparatively underexplored.

Agentic Memory Evolution (AME) refers to the continual construction, integration, and use of task-derived experiences, termed memories, as agents continuously perform tasks [11]. In this work, we further extend AME to federated scenarios and define it as *Federated AME*, where cross-agent memory exchange is used to enhance memory evolution for local tasks. However, directly introducing AME into a federated agent collaboration framework still brings *several new challenges*:

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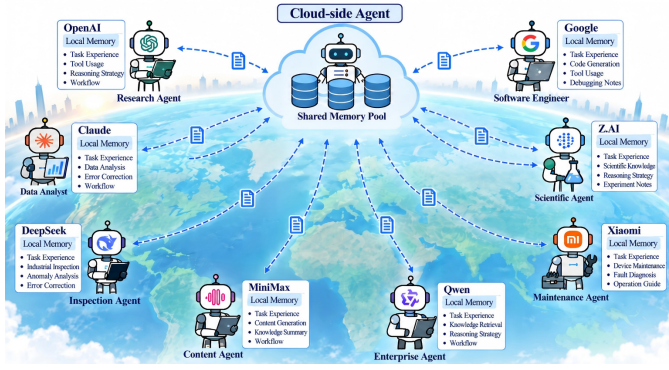


Fig. 1. Federated agent collaboration, where heterogeneous agents exchange locally acquired knowledge through a cloud-coordinated memory pool to improve local task performance. The illustrated providers include OpenAI, Anthropic, Google, DeepSeek, MiniMax, Qwen, Z.ai, and Xiaomi.

- *Wireless resource constraints.* Wireless resource constraints are often weakened or even overlooked, while naive memory-exchange paradigms may still impose pressure on limited shared bandwidth.
- *Cross-agent knowledge applicability.* Existing knowledge exchange often overlooks whether local knowledge is applicable to other agents' tasks. However, agentic memory is inherently personalized to local tasks, so indiscriminate sharing may introduce irrelevant or redundant information and consume unnecessary bandwidth.
- *Semantic quality of exchanged knowledge.* The semantic quality of exchanged knowledge is often insufficiently considered, although different memory representations may carry different amounts of useful semantic information and incur different communication payloads.

Accordingly, *the communication-efficient federated AME frameworks for wireless environments remain underexplored.*

To address these challenges, we propose **FedMEvo**, a joint agentic memory evolution and communication optimization framework for wireless federated agent collaboration. FedMEvo explicitly incorporates wireless communication constraints into federated AME and jointly optimizes on-demand cross-agent memory exchange and wireless resource allocation according to task-aware memory utility. Moreover, a multi-granularity memory representation strategy is introduced into the co-optimization, enabling finer control over the semantic information and communication payload of exchanged memories. *The Main Contributions* are summarized as follows:

- We introduce **FedMEvo**, a communication-efficient federated AME framework for wireless federated agent collaboration. It jointly optimizes task-aware agentic-memory exchange and wireless resource allocation. (See Sec. III for the complete workflow; see Fig. 2 for the overview.)
- We establish a rigorous mathematical characterization of the proposed co-optimization by deriving the unique closed-form optimal bandwidth allocation, revealing cross-agent communication coupling, and proving convergence of the discrete optimizer. (See Sec. IV for the complete mathematical analysis and proofs.)
- Extensive experiments demonstrate the superior task performance and communication efficiency of FedMEvo, as well as its practical applicability in a real-world IoT scenario. (Table II presents the complete experimental roadmap.)

II. RELATED WORK

A. Federated Agentic Collaboration

Conventional federated learning (FL) primarily supports multi-party knowledge interaction for collaborative model training and optimization [14], [15]. With the emergence of LLMs, the federated PEFT paradigm reduces trainable and exchanged parameters through lightweight adaptation [16], [17], while federated prompt learning further shifts collaborative optimization from model parameters to compact prompts [18]–[20]. Following this progression, federated agentic collaboration increasingly freezes models and exchanges reusable semantic objects, including context, skills, knowledge, and memory [3]–[6]. To further improve communication efficiency in wireless environments, recent studies investigate bandwidth-aware agentic semantic retrieval [7], self-evolving Agentic AI [1], federated Agentic AI [2], personalized edge-agent interoperability [21], and autonomous RAN agents [22]. Optimization-based wireless AI also emphasizes resource-constrained intelligent decision making [8]. However, semantic knowledge exchange and wireless resource allocation are still commonly optimized separately. FedMEvo further advances this direction by jointly optimizing federated agentic-memory evolution and wireless resource allocation, thereby reducing communication overhead while enabling more task-aware and effective cross-agent memory exchange.

B. Agentic Memory Evolution

Agentic memory organizes and reuses knowledge distilled from past interactions to support persistent reasoning and task execution, and has evolved from static storage and retrieval toward continual evolution through dynamic organization and refinement [10]–[12]. Recent studies further extend this paradigm toward cross-agent knowledge interaction through federated skill evolution [3], lightweight memory sharing [23], semantic knowledge evolution [4], and world-model federation [5]. Recent memory management methods have also begun to account for the storage and transmission costs of reusable agent knowledge [13].

TABLE I
COMPARISON OF REPRESENTATIVE AGENTIC-MEMORY EVOLUTION METHODS. ●, ○, AND ○ DENOTE EXPLICIT, PARTIAL/IMPLICIT, AND UNSUPPORTED FEATURES, RESPECTIVELY.

Work	Memory Evol.	Cross-Agent Exchange	Recipient Adaptive	Task-Aware Selection	Comm.-Cost Aware	Co-optimization
A-MEM [10]	●	○	○	●	○	○
Evo-Memory [11]	●	○	○	●	○	○
MemSkill [12]	●	○	○	●	○	○
FICAL [6]	○	●	○	○	●	○
FederatedSkill [3]	●	●	●	●	○	○
SoHip [23]	●	●	○	○	○	○
FedAgentKE [4]	●	●	●	●	○	○
FedWorld [5]	○	●	●	●	○	○
CURATOR [13]	●	●	●	●	●	○
FedMEvo	●	●	●	●	●	●

As summarized in Table I, existing methods typically focus on either memory evolution or cross-agent communication efficiency, with the two objectives treated as separate design problems. FedMEvo instead jointly addresses agentic-memory evolution and communication-cost reduction within a unified framework, as shown in Fig. 4.

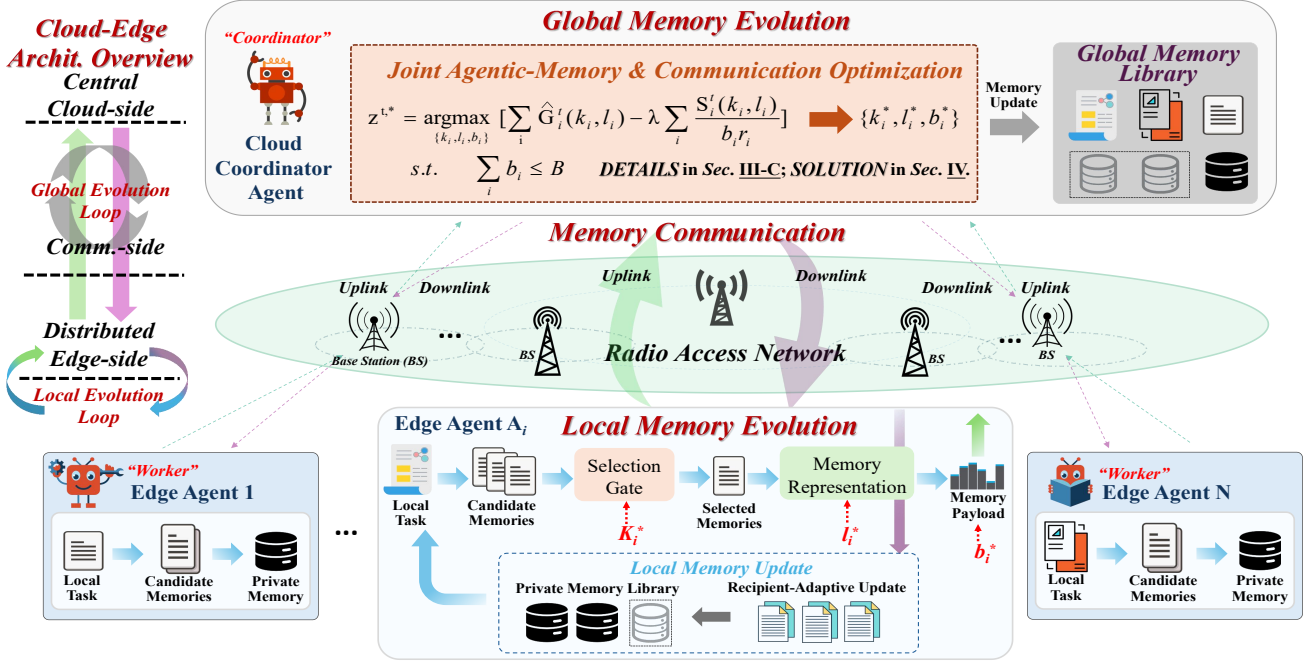


Fig. 2. Overview of FedMEvo. The Main Workflow illustrates local memory evolution, wireless memory communication, and coordinator-assisted global memory evolution. The Cloud-Edge Architecture Overview further summarizes their organization across the cloud, communication, and edge sides.

III. METHODOLOGY

A. System Abstraction and Problem Formulation

For the federated AME, the proposed FedMEvo consists of multiple edge agents that maintain private agentic-memory libraries and a coordinator that orchestrates cross-agent memory evolution and wireless resource allocation. As illustrated in Fig. 2, the overall workflow is summarized in Algorithm 1.

At evolution round t , the federation contains N edge agents. Each agent A_i maintains a local memory library $\mathcal{L}_i^t$ and constructs M_i candidate memories from its local task:

$$\mathcal{A} = \{A_1, \dots, A_N\}, \quad \mathcal{M}_i^t = \{m_{i,1}^t, \dots, m_{i,M_i}^t\}. \quad (1)$$

The coordinator maintains a shared memory pool $\mathcal{M}_g^{t-1}$ for cross-agent memory exchange.

For agent A_i , the wireless state is represented by the signal-to-noise ratio (SNR) γ_i^t , with spectral efficiency $r_i^t = \log_2(1 + \gamma_i^t)$. FedMEvo jointly considers the selected memory quantity k_i^t , representation level ℓ_i^t , and transmission bandwidth b_i^t :

$$\mathbf{z}^t = \{(k_i^t, \ell_i^t, b_i^t)\}_{i=1}^N, \quad b_i^t \geq 0, \quad \sum_{i=1}^N b_i^t \leq B, \quad (2)$$

where B denotes the total available transmission bandwidth.

The memory quantity and representation level determine the transmitted payload, while the wireless state and bandwidth determine the corresponding transmission cost, as modeled in Sec. III-B. Their joint optimization with wireless resource allocation is developed in Sec. IV. The optimized memories are then integrated into the shared memory pool and adapted to individual recipients in Sec. III-D.

B. Agentic-Memory Representation and Utility Modeling

Given the candidate memory set $\mathcal{M}_i^t$ in Eq. 1, FedMEvo characterizes each candidate in terms of semantic representation, estimated utility, and communication payload, and

organizes these candidate-level characteristics into discrete memory operating points for subsequent joint optimization.

1) *Multi-Granularity Memory Representation*. A fixed memory representation offers limited flexibility in balancing task-relevant semantic richness against communication payload. FedMEvo therefore represents each candidate memory $m_{i,j}^t$ at multiple semantic granularity levels,

$$m_{i,j}^{t,(\ell)} = \mathcal{R}_\ell(m_{i,j}^t), \quad s_{i,j}^{t,(\ell)} = \text{Size}(m_{i,j}^{t,(\ell)}), \quad \ell \in \mathcal{G}, \quad (3)$$

where $\mathcal{G}$ denotes the representation levels and $s_{i,j}^{t,(\ell)}$ is the corresponding payload size. Specifically, we adopt *Compact*, *Standard*, and *Detailed* representations, providing different semantic-detail-communication-cost operating points.

2) *Lightweight Candidate-Memory Summary*. To support efficient memory and communication decisions at the coordinator, each agent summarizes its candidate memories using compact semantic, utility, and payload descriptors. For candidate $m_{i,j}^t$, the summary is defined as

$$\mathbf{d}_{i,j}^t = \left(\bar{\mathbf{e}}_{i,j}^t, \tau_{i,j}^t, \left\{ \hat{q}_{i,j}^{t,(\ell)}, s_{i,j}^{t,(\ell)} \right\}_{\ell \in \mathcal{G}} \right), \quad (4)$$

where $\bar{\mathbf{e}}_{i,j}^t$ is the lightweight quantized preview embedding of candidate memory $m_{i,j}^t$, $\tau_{i,j}^t$ denotes its capability tag, and $\hat{q}_{i,j}^{t,(\ell)}$ denotes the locally estimated task utility under representation level ℓ . Meanwhile, agent A_i summarizes the capabilities required by its local tasks but not yet covered by its local memory library as

$$\mathcal{D}_i^t = \mathcal{T}_i^t \setminus \mathcal{C}(\mathcal{L}_i^t), \quad (5)$$

where $\mathcal{T}_i^t$ denotes the capability set relevant to the local tasks, $\mathcal{L}_i^t$ is the current local memory library, and $\mathcal{C}(\mathcal{L}_i^t)$ denotes the capabilities already covered by this library.

Accordingly, $\mathcal{D}_i^t$ provides a coarse-grained summary of the knowledge demand of agent A_i , whereas $\bar{\mathbf{e}}_{i,j}^t$ represents the fine-grained semantic content of candidate memory $m_{i,j}^t$.

Algorithm 1 FedMEvo: Federated Agentic Memory Evolution

Input: Edge agents $\mathcal{A} = \{A_1, \dots, A_N\}$, local memory libraries $\{\mathcal{L}_i^1\}_{i=1}^N$, shared memory pool $\mathcal{M}_g^0$, and total bandwidth B .
Output: Evolved local memory libraries $\{\mathcal{L}_i^t\}_{i=1}^N$.

- 1: Initialize evolution round $t \leftarrow 1$.
- 2: **repeat**
- Local candidate-memory construction*
- 3: **for** each agent A_i **do** ▷ Local memory modeling
- 4: Construct candidate memories $\mathcal{M}_i^t$ from local tasks.
- 5: Generate multi-granularity representations and descriptors via Eqs. 3–4. Derive demand $\mathcal{D}_i^t$ using Eq. 5.
- 6: **end for**
- Utility modeling and joint optimization*
- 7: Estimate collaborative memory utilities using Eqs. 6–9.
- 8: Construct memory menus $\{\Omega_i^t\}_{i=1}^N$ using Eqs. 10–11.
- 9: Solve Eq. 15 to obtain $\mathbf{z}^{t,*} = \{(k_i^{t,*}, \ell_i^{t,*}, b_i^{t,*})\}_{i=1}^N$.
- Recipient-adaptive memory evolution*
- 10: **for** each agent A_i **do** ▷ Optimized memory transmission
- 11: Upload $S_i^{t,*}$ according to Eq. 12.
- 12: **end for**
- 13: Update the shared memory pool $\mathcal{M}_g^t$ using Eq. 13.
- 14: **for** each recipient A_j **do** ▷ Recipient-specific update
- 15: Derive Δ_j^t and update $\mathcal{L}_j^{t+1}$ using Eq. 14.
- 16: **end for**
- 17: Evaluate next-round redundancy for ζ^{t+1} via Eq. 16.
- 18: $t \leftarrow t + 1$.
- 19: **until** $\zeta^t = 1$
- 20: **return** $\{\mathcal{L}_i^t\}_{i=1}^N$.

3) *Demand- and Redundancy-Aware Utility Modeling.* Based on $\mathbf{d}_{i,j}^t$ in Eq. 4 and $\{\mathcal{D}_r^t\}_{r=1}^N$ in Eq. 5, the coordinator models the collaborative utility of each candidate from task utility, cross-agent demand, and semantic redundancy.

Specifically, the redundancy of candidate memory $m_{i,j}^t$ with respect to the current shared memory pool is measured as

$$\rho_{i,j}^t = \max_{m \in \mathcal{M}_g^{t-1}} \text{sim}(\mathbf{e}_{i,j}^t, \mathbf{e}_m), \quad \rho_{i,j}^t \in [0, 1], \quad (6)$$

where a larger $\rho_{i,j}^t$ indicates stronger semantic redundancy. We further define the redundancy-aware weight as

$$\alpha_{i,j}^t = \alpha_{\min} + (1 - \alpha_{\min})(1 - \rho_{i,j}^t), \quad (7)$$

where $\alpha_{\min} > 0$ preserves the contribution of locally useful memories when semantic overlap exists.

To characterize cross-agent demand, the uncovered capabilities of the other agents are aggregated as

$$\mathcal{D}_{-i}^t = \bigcup_{r \neq i} \mathcal{D}_r^t, \quad d_{i,j}^t = \text{Match}(\tau_{i,j}^t, \mathcal{D}_{-i}^t) \in [0, 1], \quad (8)$$

where $d_{i,j}^t$ measures how well candidate memory $m_{i,j}^t$ matches the current cross-agent demand. The utility of candidate $m_{i,j}^t$ at representation level ℓ is further modeled as

$$v_{i,j}^{t,(\ell)} = \alpha_{i,j}^t \left(\left[\hat{q}_{i,j}^{t,(\ell)} \right]_+ + w_d d_{i,j}^t \right), \quad (9)$$

where w_d controls the contribution of cross-agent demand.

4) *Memory Utility–Communication Menu.* Since $v_{i,j}^{t,(\ell)}$ varies across representation levels, the candidate memories of agent A_i are ranked separately for each ℓ . Let $\pi_i^{t,(\ell)}(r)$ denote the

index of the r -th ranked candidate under representation level ℓ . Selecting the top k candidates yields the aggregated utility and payload

$$\hat{G}_i^t(k, \ell) = \sum_{r=1}^k v_{i, \pi_i^{t,(\ell)}(r)}^{t,(\ell)}, S_i^t(k, \ell) = \sum_{r=1}^k s_{i, \pi_i^{t,(\ell)}(r)}^{t,(\ell)}. \quad (10)$$

For $k = 0$, no memory is selected, and thus $\hat{G}_i^t(0, \ell) = S_i^t(0, \ell) = 0$. Accordingly, the memory-side decision space of agent A_i is organized as the utility–communication menu

$$\Omega_i^t = \left\{ \left(k, \ell, \hat{G}_i^t(k, \ell), S_i^t(k, \ell) \right) : k \in \{0, \dots, M_i\}, \ell \in \mathcal{G} \right\}. \quad (11)$$

Here, k controls the number of transmitted memories, while ℓ controls their representation detail. The resulting menus $\{\Omega_i^t\}_{i=1}^N$ provide the discrete memory-side operating points for the joint optimization developed in Sec. IV.

C. Joint Agentic-Memory and Communication Optimization

The memory menus $\{\Omega_i^t\}_{i=1}^N$ in Eq. 11 characterize the utility and payload of different memory configurations, while the wireless state r_i^t and allocated bandwidth b_i^t determine the corresponding transmission cost. FedMEvo therefore jointly determines the memory quantity, representation level, and transmission bandwidth for each agent.

Let $\mathcal{I}^t = \{i \mid k_i^t > 0\}$ denote the active-agent set, and define the joint decision as $\mathbf{z}^t = \{(k_i^t, \ell_i^t, b_i^t)\}_{i=1}^N$. The resulting joint optimization is formulated in Eq. 15, where $\lambda > 0$ controls the tradeoff between memory utility and transmission cost. Sec. IV then derives the optimal bandwidth allocation, reveals the resulting cross-agent communication coupling, and further solves the reduced discrete problem efficiently. The resulting $\mathbf{z}^{t,*}$ determines the uploaded memory payloads and supports the recipient-adaptive memory evolution in Sec. III-D.

D. Recipient-Adaptive Memory Evolution

Given the optimized decision $\mathbf{z}^{t,*} = \{(k_i^{t,*}, \ell_i^{t,*}, b_i^{t,*})\}_{i=1}^N$, agent A_i uploads

$$S_i^{t,*} = \left\{ m_{i, \pi_i^{t,(\ell_i^{t,*})}(r)}^{t,(\ell_i^{t,*})} \mid 1 \leq r \leq k_i^{t,*} \right\}, k_i^{t,*} = 0 \Rightarrow S_i^{t,*} = \emptyset. \quad (12)$$

The shared memory pool is updated as

$$\mathcal{M}_g^t = \text{Merge} \left(\mathcal{M}_g^{t-1}, \bigcup_{i=1}^N S_i^{t,*} \right). \quad (13)$$

FedMEvo adapts shared memories to each recipient using its local demand and semantic coverage. For recipient A_j , the coverage of memory m is $\kappa_j^t(m) = \max_{\tilde{m} \in \mathcal{L}_j^t} \text{sim}(\mathbf{e}_m, \mathbf{e}_{\tilde{m}})$. The recipient-specific update is then derived as

$$\Delta_j^t = \text{Select}(\mathcal{M}_g^t, \mathcal{L}_j^t; \mathcal{D}_j^t, \kappa_j^t), \mathcal{L}_j^{t+1} = \mathcal{U}(\mathcal{L}_j^t, \Delta_j^t). \quad (14)$$

Thus, the shared pool yields recipient-specific updates Δ_j^t .

The updated libraries generate the next-round candidates $\{\mathcal{M}_i^{t+1}\}_{i=1}^N$, whose redundancy is evaluated using Eq. 6. FedMEvo terminates the evolution when the remaining candidates become highly redundant:

$$\zeta^{t+1} = \text{Stop}_{\text{pred}}(\{\rho_{i,j}^{t+1}\}) \in \{0, 1\}, \begin{cases} \text{terminate,} & \zeta^{t+1} = 1, \\ t \leftarrow t + 1, & \zeta^{t+1} = 0. \end{cases} \quad (16)$$

This completes the federated agentic-memory evolution.

$$\mathbf{z}^{t,*} = \arg \max_{\mathbf{k}^t, \ell^t, \mathbf{b}^t} \sum_{i=1}^N \widehat{G}_i^t(k_i^t, \ell_i^t) - \lambda \sum_{i \in \mathcal{I}^t} \frac{S_i^t(k_i^t, \ell_i^t)}{b_i^t r_i^t} \quad (15)$$

$$\text{s.t.} \quad (k_i^t, \ell_i^t) \in \Omega_i^t, \quad \forall i, \quad b_i^t > 0, \quad i \in \mathcal{I}^t, \quad b_i^t = 0, \quad i \notin \mathcal{I}^t, \quad \sum_{i \in \mathcal{I}^t} b_i^t \leq B.$$

IV. MATHEMATICAL ANALYSIS AND EFFICIENT OPTIMIZATION

A. Optimal Bandwidth Allocation and Problem Reduction

We first analyze the continuous bandwidth variables in Eq. 15 and derive their optimal allocation for fixed memory decisions.

Assumption 1. Within each evolution round t , the memory menus $\{\Omega_i^t\}_{i=1}^N$, spectral efficiencies $\{r_i^t\}_{i=1}^N$, and total bandwidth B remain unchanged during one execution of the joint optimizer (see Eqs. 2 and 11).

For fixed (k_i^t, ℓ_i^t) , define the effective communication load of active agent A_i as

$$a_i^t \triangleq \frac{S_i^t(k_i^t, \ell_i^t)}{r_i^t}, \quad i \in \mathcal{I}^t. \quad (17)$$

The bandwidth-dependent term in Eq. 15 is reduced to

$$\min_{\{b_i^t\}} \sum_{i \in \mathcal{I}^t} \frac{a_i^t}{b_i^t}, \quad \text{s.t.} \quad b_i^t > 0, \quad \sum_{i \in \mathcal{I}^t} b_i^t \leq B. \quad (18)$$

Theorem 1. Optimal bandwidth allocation. Under Assumption 1, for fixed memory decisions, Eq. 18 admits the unique optimal allocation

$$b_i^{t,*} = B \frac{\sqrt{a_i^t}}{\sum_{j \in \mathcal{I}^t} \sqrt{a_j^t}} = B \frac{\sqrt{S_i^t(k_i^t, \ell_i^t)/r_i^t}}{\sum_{j \in \mathcal{I}^t} \sqrt{S_j^t(k_j^t, \ell_j^t)/r_j^t}}. \quad (19)$$

Proof. Fix the memory decisions $\{(k_i^t, \ell_i^t)\}_{i \in \mathcal{I}^t}$. Under Assumption 1, a_i^t in Eq. 17 remains fixed during the bandwidth optimization. Since $a_i^t > 0$ for each active agent, we have

$$\frac{\partial^2}{\partial (b_i^t)^2} \left(\frac{a_i^t}{b_i^t} \right) = \frac{2a_i^t}{(b_i^t)^3} > 0, \quad (20)$$

and hence Eq. 18 is strictly convex over $b_i^t > 0$.

Moreover,

$$\frac{\partial}{\partial b_i^t} \left(\frac{a_i^t}{b_i^t} \right) = -\frac{a_i^t}{(b_i^t)^2} < 0, \quad (21)$$

so increasing the allocated bandwidth continuously decreases the communication cost. Therefore, the optimal allocation fully utilizes the available bandwidth:

$$\sum_{i \in \mathcal{I}^t} b_i^{t,*} = B. \quad (22)$$

Using Eq. 22, the Lagrangian of Eq. 18 is

$$\mathcal{L} = \sum_{i \in \mathcal{I}^t} \frac{a_i^t}{b_i^t} + \mu \left(\sum_{i \in \mathcal{I}^t} b_i^t - B \right). \quad (23)$$

The KKT stationarity condition gives

$$\frac{\partial \mathcal{L}}{\partial b_i^t} = -\frac{a_i^t}{(b_i^t)^2} + \mu = 0, \quad (24)$$

which yields

$$b_i^t = \sqrt{\frac{a_i^t}{\mu}}. \quad (25)$$

Substituting Eq. 25 into Eq. 22, we obtain

$$\frac{1}{\sqrt{\mu}} \sum_{j \in \mathcal{I}^t} \sqrt{a_j^t} = B, \quad \sqrt{\mu} = \frac{\sum_{j \in \mathcal{I}^t} \sqrt{a_j^t}}{B}. \quad (26)$$

Finally, substituting Eq. 26 into Eq. 25 gives Eq. 19. ■

Theorem 1 shows that the allocated bandwidth scales with $\sqrt{S_i^t/r_i^t}$, thereby assigning more bandwidth to agents with larger payloads or lower spectral efficiency. Substituting Eq. 19 into the communication term gives

$$C^t(\mathbf{k}^t, \ell^t) = \sum_{i \in \mathcal{I}^t} \frac{S_i^t(k_i^t, \ell_i^t)}{b_i^{t,*} r_i^t} = \frac{1}{B} \left(\sum_{i \in \mathcal{I}^t} \sqrt{\frac{S_i^t(k_i^t, \ell_i^t)}{r_i^t}} \right)^2. \quad (27)$$

Consequently, Eq. 15 is reduced to

$$\max_{\mathbf{k}^t, \ell^t} \Phi^t = \sum_{i=1}^N \widehat{G}_i^t(k_i^t, \ell_i^t) - \lambda C^t(\mathbf{k}^t, \ell^t), \quad \text{s.t.} \quad (k_i^t, \ell_i^t) \in \Omega_i^t. \quad (28)$$

The continuous bandwidth variables are thus eliminated, while the resulting communication cost remains coupled across active agents.

B. Cross-Agent Communication Coupling

We next characterize the coupling induced by the reduced communication cost in Eq. 27.

Lemma 1. Non-separable communication cost. Under Assumption 1, the reduced communication cost admits the exact decomposition

$$C^t = \frac{1}{B} \left[\sum_{i \in \mathcal{I}^t} a_i^t + 2 \sum_{i,j \in \mathcal{I}^t, i < j} \sqrt{a_i^t a_j^t} \right]. \quad (29)$$

Hence, C^t is non-separable across active agents due to the cross-agent terms $\sqrt{a_i^t a_j^t}$.

Proof. By Eq. 17, Eq. 27 can be rewritten as

$$C^t = \frac{1}{B} \left(\sum_{i \in \mathcal{I}^t} \sqrt{a_i^t} \right)^2. \quad (30)$$

Expanding the squared sum gives

$$\left(\sum_{i \in \mathcal{I}^t} \sqrt{a_i^t} \right)^2 = \sum_{i \in \mathcal{I}^t} a_i^t + 2 \sum_{i,j \in \mathcal{I}^t, i < j} \sqrt{a_i^t a_j^t}. \quad (31)$$

Substituting Eq. 31 into Eq. 30 yields Eq. 29. Since each cross term depends on two active agents, C^t cannot be decomposed into independent per-agent communication costs. ■

Lemma 1 shows that eliminating the continuous bandwidth variables does not eliminate cross-agent interaction. We further quantify this coupling through marginal communication cost.

Proposition 1. *Cross-agent marginal communication coupling. Consider changing the memory configuration of active agent A_i from (k_i^t, ℓ_i^t) to $(\tilde{k}_i^t, \tilde{\ell}_i^t)$, and define*

$$\tilde{a}_i^t \triangleq \frac{S_i^t(\tilde{k}_i^t, \tilde{\ell}_i^t)}{r_i^t}, \quad z_{-i}^t \triangleq \sum_{\substack{j \in \mathcal{I}^t \\ j \neq i}} \sqrt{a_j^t}. \quad (32)$$

Then the resulting marginal communication cost is

$$\Delta C_i^t = \frac{1}{B} \left[\tilde{a}_i^t - a_i^t + 2z_{-i}^t \left(\sqrt{\tilde{a}_i^t} - \sqrt{a_i^t} \right) \right]. \quad (33)$$

Proof. From Eq. 30, separate the contribution of agent A_i from those of the remaining active agents:

$$C^t = \frac{1}{B} \left(\sqrt{a_i^t} + z_{-i}^t \right)^2. \quad (34)$$

After replacing the local configuration of A_i , the communication cost becomes

$$\tilde{C}_i^t = \frac{1}{B} \left(\sqrt{\tilde{a}_i^t} + z_{-i}^t \right)^2. \quad (35)$$

Therefore,

$$\begin{aligned} \Delta C_i^t &\triangleq \tilde{C}_i^t - C^t \\ &= \frac{1}{B} \left[\tilde{a}_i^t - a_i^t + 2z_{-i}^t \left(\sqrt{\tilde{a}_i^t} - \sqrt{a_i^t} \right) \right], \end{aligned} \quad (36)$$

which is exactly Eq. 33. Since z_{-i}^t aggregates the effective loads of all other active agents, the marginal communication cost of a local memory change depends on the concurrent communication state of the federation. ■

Proposition 1 establishes the cross-agent coupling between memory decisions and communication cost. Substituting Eq. 27 into Eq. 28 leaves a discrete coupled optimization, which is addressed next.

C. Efficient Discrete Optimization

After eliminating the continuous bandwidth variables, we solve the remaining discrete problem in Eq. 28. Define the local memory-decision set as the projection of Ω_i^t :

$$\mathcal{X}_i^t = \left\{ (k, \ell) \mid (k, \ell, \hat{G}_i^t, S_i^t) \in \Omega_i^t \right\}, \quad \mathcal{X}^t = \prod_{i=1}^N \mathcal{X}_i^t. \quad (37)$$

Let $\mathbf{x}^t = \{x_i^t\}_{i=1}^N$ with $x_i^t = (k_i^t, \ell_i^t)$. Then Eq. 28 can be written compactly as

$$\max_{\mathbf{x}^t \in \mathcal{X}^t} \Phi^t(\mathbf{x}^t) = \sum_{i=1}^N \hat{G}_i^t(x_i^t) - \lambda C^t(\mathbf{x}^t). \quad (38)$$

Direct enumeration over $\mathcal{X}^t$ grows combinatorially with the number of agents and memory configurations. We therefore optimize Eq. 38 through incremental memory operations. Let the ordered granularity set be $\mathcal{G} = \{\ell_1, \dots, \ell_L\}$. For agent A_i , we consider

$$\begin{aligned} A_i : (k_i^t, \ell_q) &\mapsto (k_i^t + 1, \ell_q), \\ R_i : (k_i^t, \ell_q) &\mapsto (k_i^t, \ell_{q+1}), \end{aligned} \quad (39)$$

where A_i adds one ranked memory and R_i refines its representation level. Denoting the state transition induced by operation u as $\mathcal{T}_u(\mathbf{x})$, the feasible operation set is

$$\mathcal{U}(\mathbf{x}) = \{u \in \{A_i, R_i\}_{i=1}^N \mid \mathcal{T}_u(\mathbf{x}) \in \mathcal{X}^t\}. \quad (40)$$

For each feasible operation, its marginal objective gain is

$$\begin{aligned} \Delta \Phi^t(u \mid \mathbf{x}) &\triangleq \Phi^t(\mathcal{T}_u(\mathbf{x})) - \Phi^t(\mathbf{x}) \\ &= \Delta \hat{G}^t(u \mid \mathbf{x}) - \lambda \Delta C^t(u \mid \mathbf{x}), \end{aligned} \quad (41)$$

where the communication increment $\Delta C^t(u \mid \mathbf{x})$ follows from Eq. 27 and retains the cross-agent coupling established in Proposition 1. At iteration h , FedMEvo selects

$$u^{(h),*} = \arg \max_{u \in \mathcal{U}(\mathbf{x}^{(h)})} \Delta \Phi^t(u \mid \mathbf{x}^{(h)}), \quad (42)$$

and updates

$$\mathbf{x}^{(h+1)} = \begin{cases} \mathcal{T}_{u^{(h),*}}(\mathbf{x}^{(h)}), & \Delta \Phi^t(u^{(h),*} \mid \mathbf{x}^{(h)}) > 0, \\ \mathbf{x}^{(h)}, & \text{otherwise.} \end{cases} \quad (43)$$

Lemma 2. *Strict ascent property. Every accepted operation in Eq. 43 strictly increases the reduced objective, i.e.,*

$$\Phi^t(\mathbf{x}^{(h+1)}) > \Phi^t(\mathbf{x}^{(h)}). \quad (44)$$

Proof. An operation is accepted only if $\Delta \Phi^t(u^{(h),*} \mid \mathbf{x}^{(h)}) > 0$ in Eq. 43. By the definition in Eq. 41,

$$\Phi^t(\mathbf{x}^{(h+1)}) - \Phi^t(\mathbf{x}^{(h)}) = \Delta \Phi^t(u^{(h),*} \mid \mathbf{x}^{(h)}) > 0, \quad (45)$$

which directly yields Eq. 44. ■

Theorem 2. *Finite convergence. Starting from any feasible $\mathbf{x}^{(0)} \in \mathcal{X}^t$, the update rule in Eq. 43 terminates after a finite number of accepted operations.*

Proof. From Eq. 11, each agent has a finite number of memory quantities and representation levels; hence

$$|\mathcal{X}^t| = \prod_{i=1}^N |\mathcal{X}_i^t| < \infty. \quad (46)$$

By Lemma 2, each accepted operation increases $\Phi^t(\mathbf{x})$, so an accepted update cannot revisit a previously visited state with the same objective value. Since the feasible state in Eq. 46 is finite, only finitely many strictly improving transitions can occur. Therefore, Eq. 43 terminates in finite steps. ■

Corollary 1. *One-move local optimality. Let $\mathbf{x}^{t,*}$ denote the state at termination. Then*

$$\max_{u \in \mathcal{U}(\mathbf{x}^{t,*})} \Delta \Phi^t(u \mid \mathbf{x}^{t,*}) \leq 0. \quad (47)$$

Proof. If there existed a feasible operation u satisfying $\Delta \Phi^t(u \mid \mathbf{x}^{t,*}) > 0$, then Eq. 42 would select a positive-gain operation and Eq. 43 would perform another update, contradicting termination. Hence Eq. 47 holds. ■

Starting from the all-idle configuration at the lowest representation level, each agent admits at most M_i memory-addition operations and $L - 1$ representation refinements. Hence, the number of accepted operations satisfies

$$H^t \leq \sum_{i=1}^N (M_i + L - 1). \quad (48)$$

With incremental evaluation of Eq. 41, at most $2N$ feasible operations are examined per iteration, yielding the computational complexity

$$\mathcal{O}(NH^t) = \mathcal{O}\left(N \sum_{i=1}^N (M_i + L)\right). \quad (49)$$

Thus, the coupled problem can be solved without exhaustive enumeration of the Cartesian Decision Space.

TABLE II
EXPERIMENTAL ROADMAP ACROSS FOUR RESEARCH QUESTIONS.

Research Goal	Experimental Design & Evaluation	Key Results Summary
RQ1: Performance–Communication Effectiveness	• Comparative Experiments: Comparing overall performance across benchmarks (Table III), model settings (Table VI), and low-communication baselines (Fig. 4).	Superior overall task performance.
	• Communication-Efficiency Analysis: Examining overall communication cost (Table III), traffic decomposition (Fig. 3), and task-wise communication composition (Fig. 11).	Communication-efficient exchange.
	• Performance–Communication Analysis: Examining task-level operating points (Fig. 5), cross-task changes (Fig. 4), configuration sensitivity (Fig. 6), and budget-constrained utility preservation (Fig. 10).	Favorable performance–communication balance.
RQ2: Co-optimization Effectiveness	• Core Component Ablation: Quantifying the contributions of memory-evolution and joint-optimization components to task performance and communication cost (Table IV).	Effective key components.
	• Agentic-Memory Evolution Analysis: Examining adaptive semantic granularity (Fig. 8) and the effectiveness of cross-client memory transfer (Fig. 9).	Effective memory evolution.
	• Joint-Optimization Analysis: Comparing alternative joint-optimization mechanisms (Table V), configuration sensitivity (Fig. 6), and utility preservation under communication budgets (Fig. 10).	Effective joint optimization.
RQ3: Robustness and Adaptability	• Task and Model Robustness: Assessing robustness across heterogeneous tasks (Table III) and homogeneous/heterogeneous backbone configurations (Table VI).	Robust across tasks and models.
	• Scalability and Data Heterogeneity: Evaluating task performance and communication behavior under varying client scales and non-IID data distributions (Fig. 7).	Robust to scale and data heterogeneity.
	• Communication Adaptability: Examining round budgets and adaptive stopping (Table VII; Fig. 12), communication-budget variation (Fig. 10), and wireless conditions (Fig. 13).	Adaptive to communication conditions.
RQ4: Real-world Application	• IoMT Methane-Risk Prediction: Evaluating methane-risk prediction effectiveness and communication cost in the industrial IoMT case study (Table IX).	Effective methane-risk prediction.
	• Edge-node Deployment Adaptability: Assessing performance and communication across edge-node model deployments (Table IX) and their performance–communication robustness (Fig. 14).	Stable across model deployments.
	• Federated Agentic Collaboration: Examining progressive agentic-memory evolution and communication growth (Fig. 15) and performance–communication robustness across edge-node model settings (Fig. 14).	Communication-efficient collaboration.

V. EXPERIMENTS & ANALYSIS

In this section, we conduct extensive experiments to evaluate the performance of FedMemOpt. *Comparative Experiments* in Sec. V-B evaluate task performance and communication efficiency against representative federated agentic baselines and structural variants; *Ablation Studies* in Sec. V-C examine the key agentic-memory evolution and communication-optimization mechanisms; *Analytical Experiments* in Sec. V-D investigate robustness, agentic-memory effectiveness, and communication-efficient joint optimization; and *Case Studies* assess practical applicability through distributed methane-risk identification in industrial IoT. Table II summarizes the experimental designs and evaluation objectives, while Sec. V-D provides an *Analytical Roadmap*.

A. Experimental Setup

1) *Experimental Goals and Roadmap:* We evaluate FedMemOpt along the following four research dimensions:

- **RQ1 (Performance–Communication Effectiveness):** Does FedMemOpt outperform representative federated agentic baselines and structural variants in task performance and communication efficiency while achieving a favorable performance–communication trade-off?
- **RQ2 (Co-optimization Effectiveness):** How do agentic-memory evolution and communication optimization jointly affect task performance and communication cost?
- **RQ3 (Robustness and Adaptability):** How robust is FedMemOpt across tasks, LLM backbones, client scales, model and data heterogeneity, and application settings?

- **RQ4 (Real-world Application):** Can FedMemOpt support communication-efficient federated agentic collaboration in practical industrial IoT scenarios?

Table II maps each research question to its corresponding experiments, evaluation objectives, and key results.

2) *Implementation Details:* For fair comparison, we follow the implementations and evaluation protocols of representative federated agentic methods, including FederatedSkill and FedWorld [3], [5]. Agentic-memory evolution task keeps all backbone models frozen and requires only a small number of rounds. In addition, all methods operate under the same reliable network setting, where packet loss and additional network jitter are excluded. Clients run GPU computation on NVIDIA RTX A6000 platforms, while the central server performs multithreaded cloud processing on a 32-core Intel Xeon Platinum 8358P CPU. Detailed hyperparameters and implementation settings are provided on Project Page.

3) *Baselines & Benchmarks:* We evaluate FedMemOpt on five benchmarks: WCNS, WCMSA, and WCHW from WirelessBench [25], ORANBench [26], and 3GPP-TSG [27]. The baselines cover representative federated agentic paradigms, including skill federation (FS-Compatible [3]), memory curation (Curator [13]), iterative ICL (FedICL [24]), world-model sharing (FedWorld [5]), semantic knowledge exchange (FedAgenticKE [4]), and federated contextual learning (FICAL [6]). Within each benchmark, all methods follow the same evaluation protocol and communication-accounting scheme.

TABLE III

MAIN COMPARISON ON FIVE WIRELESS-AGENT BENCHMARKS. EACH BENCHMARK REPORTS **TASK SCORE** ($\uparrow$) AND **COMMUNICATION** (**COMM./KB**, $\downarrow$), WHILE **TOTAL** DENOTES THE ARITHMETIC MEAN ACROSS BENCHMARKS. BEST RESULTS ARE BOLD AND SECOND-BEST RESULTS ARE UNDERLINED.

Method	WCNS		WCMSA		WCHW		ORANBench		3GPP-TSG		Total	
	Score $\uparrow$	Comm./KB $\downarrow$	Score $\uparrow$	Comm./KB $\downarrow$	Score $\uparrow$	Comm./KB $\downarrow$	Score $\uparrow$	Comm./KB $\downarrow$	Score $\uparrow$	Comm./KB $\downarrow$	Score $\uparrow$	Comm./KB $\downarrow$
<i>Baseline Methods</i>												
FS-Compatible [3]	0.396	61.8	0.514	76.3	0.184	48.0	0.607	69.0	0.200	52.5	0.380	<u>61.5</u>
Curator [13]	0.453	37.0	<u>0.587</u>	89.2	0.213	97.6	<u>0.633</u>	61.4	0.210	48.5	<u>0.419</u>	66.7
FedICL [24]	0.426	55.2	0.431	103.0	0.187	142.1	0.593	146.8	0.230	52.9	0.373	100.0
FedWorld [5]	0.483	82.5	0.456	<u>73.0</u>	0.184	79.5	0.573	81.3	0.250	74.1	0.389	78.1
FedAgentKE [4]	0.426	50.8	0.431	93.2	0.187	124.8	0.593	132.8	0.230	<u>50.3</u>	0.373	90.4
FICAL [6]	<u>0.506</u>	79.6	0.488	264.0	0.182	199.5	0.600	184.0	0.220	57.3	0.399	156.9
<i>Structural Variants</i>												
NoMemory	0.422	-	0.432	-	0.197	-	<u>0.633</u>	-	<u>0.260</u>	-	0.389	-
Local	0.432	-	0.533	-	0.178	-	0.587	-	<u>0.260</u>	-	0.398	-
FullUpload	0.398	286.4	0.524	194.6	<u>0.218</u>	328.4	0.573	391.1	0.220	410.2	0.387	322.1
FedMemOpt	0.778	<u>38.4</u>	0.765	50.0	0.291	<u>78.7</u>	0.687	50.9	0.430	64.3	0.590	56.5

Notes: KB denotes 10^3 serialized bytes. *NoMemory* uses no memory, *Local* uses only private local memory, and *FullUpload* shares the complete memory pool without the proposed optimization. *NoMemory* and *Local* incur zero communication by design and are therefore excluded from communication ranking.

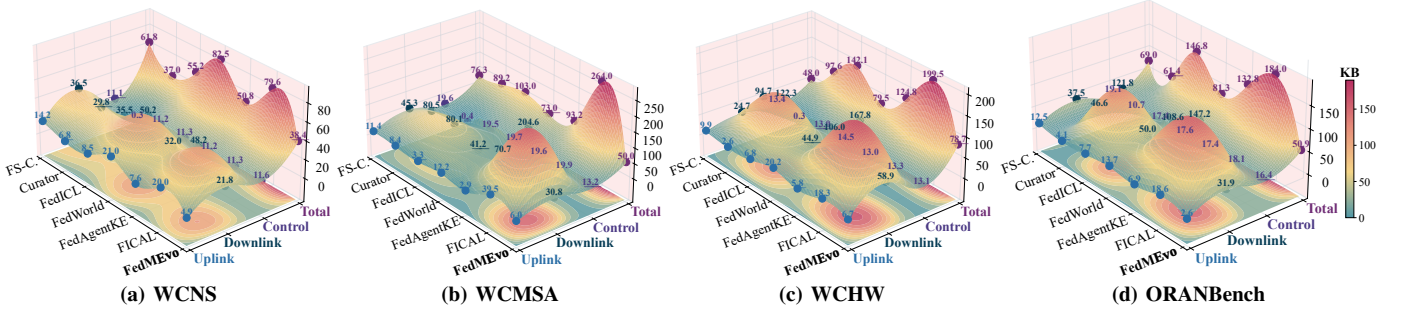


Fig. 3. Communication-specific comparison across external baselines and FedMemOpt. Markers report exact uplink, recipient-charged downlink, control, and total communication in KB. The lowest communication cost is highlighted in bold, and the second-best result is underlined.

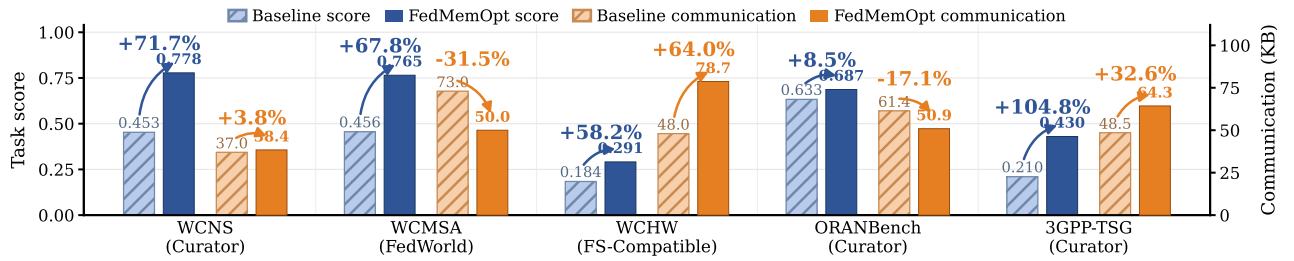


Fig. 4. Comparison between FedMemOpt and selected low-communication baselines. Hatched and solid bars denote the baselines and FedMemOpt, respectively; curved arrows indicate relative changes, with negative communication values denoting traffic reduction.

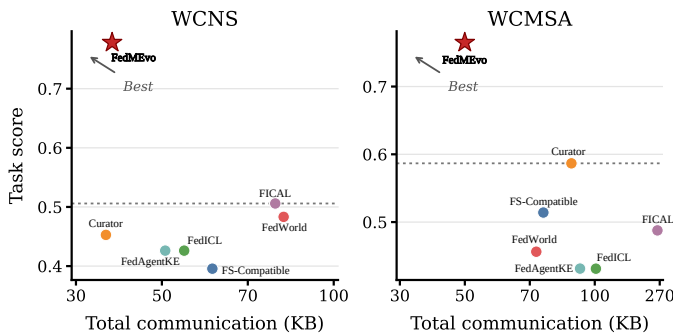


Fig. 5. Performance-communication robustness of FedMemOpt under different edge-node model settings in the IOMT system. FedMemOpt maintains favorable operating points across multiple model deployments; the upper-left direction indicates higher task performance and lower communication cost.

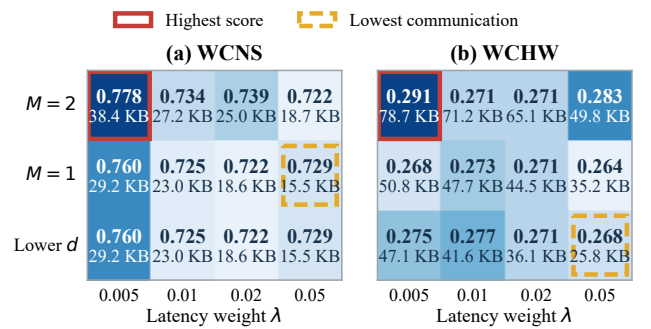


Fig. 6. Hyperparameter sensitivity analysis. Each cell reports task score and total communication; red solid and gold dashed outlines indicate the best-score and lowest-communication settings, respectively.

4) *Data Construction & Metrics*: For each benchmark, 60 development instances are distributed across clients for local agentic-memory evolution and subsequent federated collaboration. Performance is evaluated on the partitioned benchmark test set. *Task Score* follows the native metric of each benchmark, using tolerance-aware or weighted scores for WirelessBench and accuracy for the remaining benchmarks. *Communication* includes serialized uplink, recipient-charged downlink, and control traffic in 10^3 bytes. More detailed construction settings are provided on the [Project Page](#).

B. Comparative Experiments

We compare FedMemOpt with representative federated agentic baselines and structural variants in terms of **Task Performance**, **Communication Efficiency**, and **Performance–Communication Trade-offs**.

1) *Main Quantitative Results*: Table III reports the overall results across five benchmarks. FedMemOpt achieves the highest task score on all benchmarks, outperforming the strongest baselines result by 53.8% and 65.4% on WCNS and 3GPP-TSG, respectively. Meanwhile, it reduces the average communication cost by 82.5% compared with FullUpload, demonstrating *strong task performance with substantially lower communication overhead*.

2) *Communication-specific Comparison*: Fig. 3 decomposes the communication cost into uplink, downlink, and control traffic. FedMemOpt maintains competitive total communication across four benchmarks and achieves the lowest positive communication on WCMSA. Its uplink remains consistently small, while recipient-specific downlink constitutes the dominant task-dependent cost, showing that *FedMemOpt limits source-side communication through selective memory exchange while prioritizing useful recipient-specific delivery*.

3) *Performance–Communication Balance*: Fig. 4 compares FedMemOpt with representative low-communication baselines across all five benchmarks, showing consistent task-performance gains, with communication reductions on WCMSA and ORANBench and moderate communication increases on the remaining tasks. Fig. 5 further visualizes the operating points on WCNS and WCMSA, where FedMemOpt occupies favorable upper-left regions with higher task scores and competitive communication costs. These results show that *FedMemOpt improves the performance–communication operating point rather than simply trading additional communication for higher task performance*.

C. Ablation Studies

We evaluate the key designs of FedMemOpt from three perspectives: **Core Component Ablation**, **Alternative Joint-Optimization Comparison**, and **Hyperparameter Sensitivity**, covering the key mechanisms for agentic-memory evolution and communication optimization.

1) *Core Component Ablation*: Table IV evaluates the key components and structural variants of FedMemOpt. For *Memory Optimization*, all components contribute to task performance; for example, removing multi-granularity representation reduces the task score by 5.8%. For *Joint Optimization*, removing key components leads to performance drops of 5.8%–10.8%. Compared with the structural variants, FedMemOpt

TABLE IV
STRUCTURAL AND COMPONENT ABLATIONS OF FEDMEMOPT ON ORANBENCH-LITE.

Category	Variant	Task Score $\uparrow$	Communication/KB $\downarrow$			
			Up.	Down.	Ctrl.	Total
Memory Optimization	w/o Memory Evolution	0.653	5.9	30.2	9.5	45.6
	w/o Multi-granularity	0.647	7.5	36.7	13.5	57.7
	w/o Recipient Awareness	0.653	6.0	40.9	13.5	60.4
Joint Optimization	w/o Demand Awareness	0.647	3.9	21.1	12.9	37.9
	w/o Novelty Awareness	0.633	7.3	39.2	13.5	60.0
	w/o Wireless Awareness	0.613	2.9	15.5	13.5	31.9
Structural Variants	NoMemory	0.540	—	—	—	—
	Local	0.587	—	—	—	—
	FullUpload	0.573	199.0	166.2	25.9	391.1
FedMemOpt		0.687	2.6	31.9	16.4	50.9

Notes: Task Score denotes test-set accuracy, and KB denotes 10^3 serialized bytes. For memory optimization, *w/o Memory Evolution* removes iterative memory evolution; *w/o Multi-granularity* replaces multi-granularity memory representation with a single granularity; *w/o Recipient Awareness* removes recipient-specific memory delivery.

TABLE V
COMPARISON OF FEDMEMOPT WITH REPRESENTATIVE JOINT-OPTIMIZATION MECHANISMS ON ORANBENCH-LITE.

Method	Score $\uparrow$	Uplink $\downarrow$	Downlink $\downarrow$	Total $\downarrow$
OJCTA [28]	0.673	12.1	90.5	119.0
e-HARBOR [29]	0.693	4.0	48.5	68.9
DTAC [30]	0.680	3.6	44.0	64.0
CAMART [31]	0.700	3.6	46.1	66.0
FedMemOpt	0.720	2.5	27.2	46.6

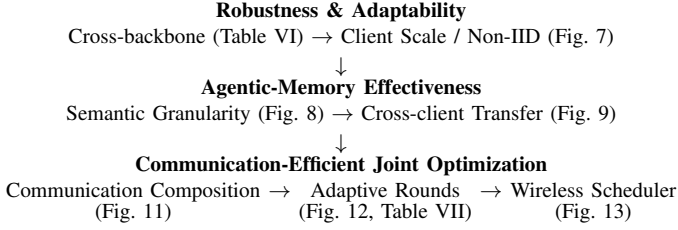
improves task performance by 17.0%–27.2% and reduces communication by 87.0% relative to FullUpload, validating the effectiveness of the proposed agentic-memory evolution and joint-optimization mechanisms.

2) *Alternative Joint-Optimization Mechanisms*: Table V compares FedMemOpt with representative alternative joint-optimization mechanisms [28]–[31] (from 2025–2026 JSAC/TMC) on ORANBench-Lite. FedMemOpt achieves the highest task score and the lowest communication cost, improving performance by 2.9% over the strongest-performing alternative while reducing communication by 27.2% relative to the lowest-communication alternative. These results validate the effectiveness of the proposed joint optimization.

3) *Hyperparameter Sensitivity*: Fig. 6 evaluates multiple configurations on two benchmarks by varying the latency weight λ , memory capacity M , and demand level. Both tasks achieve their highest scores at $M = 2$. Compared with these best-score settings, the lowest-communication configurations reduce traffic by 59.6% and 67.2% with score drops of only 6.3% and 7.9% on WCNS and WCHW, respectively. Other configurations provide intermediate operating points, demonstrating controllable performance–communication adjustment.

D. Analytical Experiments

We organize the analytical experiments into three progressive stages. We first examine **Robustness & Adaptability** under diverse configurations, then investigate **Agentic-Memory Effectiveness** through semantic granularity and cross-client transfer, and finally analyze **Communication-Efficient Joint Optimization** under wireless constraints. The *Analytical Roadmap* as shown below:



(1) *Robustness and Adaptability*. We first examine whether FedMemOpt remains effective under changes in model configurations and federated operating conditions.

Cross-backbone and Model Heterogeneity: Table VI evaluates FedMemOpt under different client-model configurations on WCMSA and ORANBench. Across the six homogeneous settings, FedMemOpt achieves the best task performance, with a maximum relative gain of 30.3%, while consistently attaining the lowest positive communication. Under heterogeneous models, it retains the lowest positive communication on both tasks while maintaining strong task performance, demonstrating robustness to backbone variations and model heterogeneity.

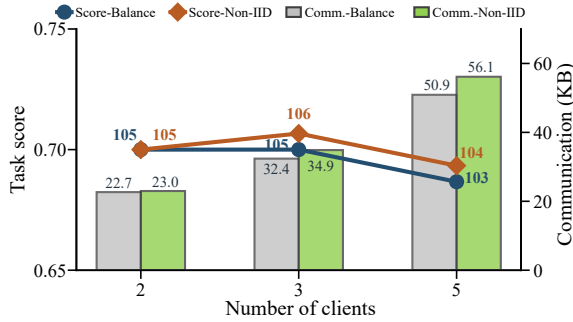


Fig. 7. Scalability and data-heterogeneity analysis on ORANBench. Lines show task scores, and bars show total communication.

Client Scalability and Data Heterogeneity: Fig. 7 evaluates FedMemOpt under varying client scales and data distributions. As the number of clients increases, score variation remains within 2.9%, while performance stays stable under both balanced and non-IID distributions. Communication increases only moderately with federation scale, demonstrating robustness to client scalability and data heterogeneity.

(2) *Agentic-Memory Effectiveness*. We evaluate the performance of agentic memory in terms of *Semantic Granularity* and *Cross-client Transfer*.

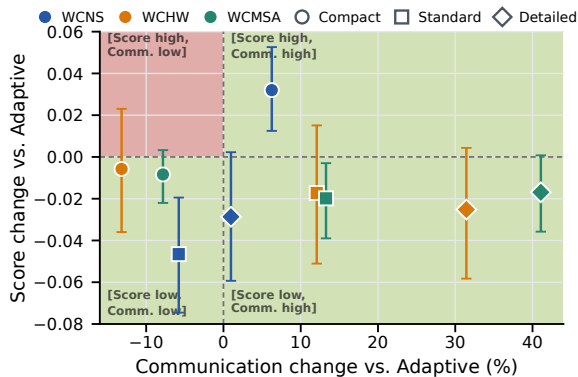


Fig. 8. Semantic-granularity analysis. Fixed-granularity variants are measured relative to Adaptive, with paired-bootstrap 95% intervals. The red upper-left region indicates score improvement and communication reduction.

Semantic Granularity Analysis: As shown in Fig. 8, adaptive granularity achieves favorable performance-communication

operating points across the benchmarks. Compared with fixed Standard granularity, it yields significant score gains on WCNS and WCMSA while reducing communication by 11.7% on WCMSA. On WCHW, adaptive granularity achieves the highest score while reducing communication by 10.8%–23.9% relative to fixed alternatives, validating the effectiveness of adaptive memory representation.

Cross-client Memory Transfer Analysis: Fig. 9 evaluates cross-client memory transfer using NoMemory as the reference. Across different settings, beneficial prediction transitions substantially outnumber harmful ones, indicating that transferred memories more often correct erroneous predictions than disrupt correct ones.

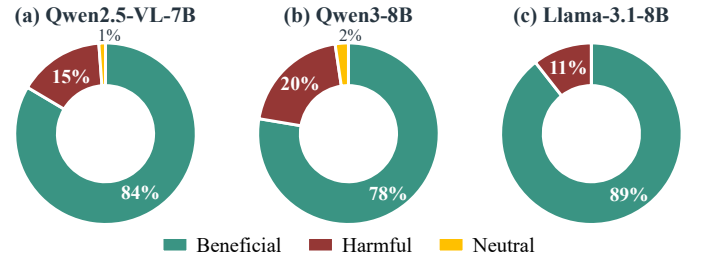


Fig. 9. Cross-client memory transfer analysis across backbone models. *Beneficial*, *harmful*, and *neutral* denote incorrect-to-correct, correct-to-incorrect, and unchanged predictions relative to NoMemory, respectively. Beneficial transitions consistently dominate harmful ones.

(3) *Communication-Efficient Joint Optimization*. We evaluate how FedMemOpt preserves task utility and controls communication under wireless constraints through *Communication-Constrained Utility Preservation*, *Communication Cost Composition*, *Dynamic Round Control*, and *Wireless Scheduling*.

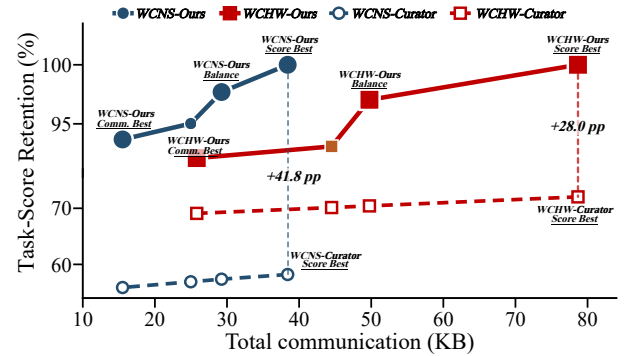


Fig. 10. Performance-communication trajectories of FedMemOpt (solid) and budget-matched Curator estimates (dashed). Vertical links indicate the retention gap at Score-Best budget.

Communication-Constrained Utility Preservation: As shown in Fig. 10, FedMemOpt retains 93.7%–100% and 92.1%–100% of its best performance across communication budgets on WCNS and WCHW, respectively. Compared with budget-matched Curator references, the retention gains reach 41.8 and 28.0 points at the Score-Best budgets, showing strong utility preservation under constrained communication.

Communication Cost Composition: As shown in Fig. 11, compared with the communication-efficient baseline FedWorld, FedMemOpt incurs lower communication costs across all evaluated tasks and communication components. The reductions are consistent across uplink, downlink, and control traffic. These results also provide a basis for the subsequent scheduling optimization.

TABLE VI
CROSS-BACKBONE AND MODEL-HETEROGENEITY COMPARISON ON WCMSA AND ORANBENCH.

Method	WCMSA								ORANBench							
	Qwen2.5-VL-7B		Qwen3-8B		Llama-3.1-8B		Heterogeneous		Qwen2.5-VL-7B		Qwen3-8B		Llama-3.1-8B		Heterogeneous	
	Score ↑	Comm. ↓	Score ↑	Comm. ↓	Score ↑	Comm. ↓	Score ↑	Comm. ↓	Score ↑	Comm. ↓	Score ↑	Comm. ↓	Score ↑	Comm. ↓	Score ↑	Comm. ↓
FS-Compatible [3]	0.514	76.3	0.596	48.0	0.556	67.0	0.469	58.7	0.607	69.0	0.713	65.6	0.660	63.8	0.640	66.5
Curator [13]	0.587	89.2	0.591	43.5	0.621	94.0	0.507	59.0	0.633	61.4	0.713	67.1	0.680	66.5	0.667	0.0
FedICL [24]	0.431	103.0	0.546	74.3	0.539	85.4	0.446	108.7	0.593	146.8	0.720	152.5	0.680	141.9	0.627	148.4
FedWorld [5]	0.456	73.0	0.561	53.0	0.548	64.0	0.451	68.3	0.573	81.3	0.700	81.8	0.640	78.9	0.627	83.7
FedAgentKE [4]	0.431	93.2	0.551	70.2	0.508	79.7	0.438	79.7	0.593	132.8	0.720	138.5	0.680	127.9	0.627	134.4
FICAL [6]	0.488	264.0	0.572	169.0	0.527	225.4	0.452	221.9	0.600	184.0	0.720	187.0	0.653	178.6	0.647	180.6
FullUpload	0.524	194.6	0.595	106.7	0.566	168.5	0.471	146.6	0.573	391.1	0.687	399.1	0.653	382.6	0.627	401.6
FedMemOpt	0.765	50.0	0.649	37.2	0.771	32.4	<u>0.476</u>	40.9	0.687	50.9	0.720	46.6	0.687	35.9	0.673	44.6

Notes: *Heterogeneous* uses a fixed client-model composition with two Qwen2.5-VL-7B-Instruct clients, two Qwen3-8B clients, and one Llama-3.1-8B-Instruct client. Best results are bold and second-best results are underlined.

TABLE VII
COMMUNICATION COST UNDER BUDGET CONSTRAINTS AND ADAPTIVE ROUND CONTROL ON 3GPP-TSG.

Budget	Dynamic	$R_{\max}$	KB ↓
Default	No	2	77.5
Unconstrained	No	2	93.2
Default	No	3	89.8
Unconstrained	No	3	124.7
Unconstrained	No	5	207.6
Unconstrained	Yes	5	207.6
Default	Yes	5	73.3

Notes: *Default* denotes the budget-constrained setting under the proposed joint-optimization mechanism; *Unconstrained* removes the pre-selection budget constraint; *Dynamic* indicates whether adaptive round control is enabled; and $R_{\max}$ denotes the maximum number of communication rounds. KB denotes 10^3 serialized bytes.

Dynamic Round Control: Table VII quantifies communication costs under different round budgets and adaptive-stopping settings, where the dynamic default policy achieves 73.3 KB of communication. Fig. 12 further visualizes the cumulative communication trajectories: the default policy stops after four rounds as candidate memories become highly redundant, whereas unconstrained execution continues to $R_{\max} = 5$ and increases communication by 183.2%. These results demonstrate the effectiveness of redundancy-aware adaptive stopping in suppressing unnecessary late-round communication.

Wireless Scheduling and Near-optimality: Finally, we examine the quality of wireless-aware joint scheduling under different bandwidth-latency conditions. As shown in Fig. 13, FedMemOpt reaches the optimal objective in 17 of the 20 evaluated settings, while the remaining cases exhibit only small

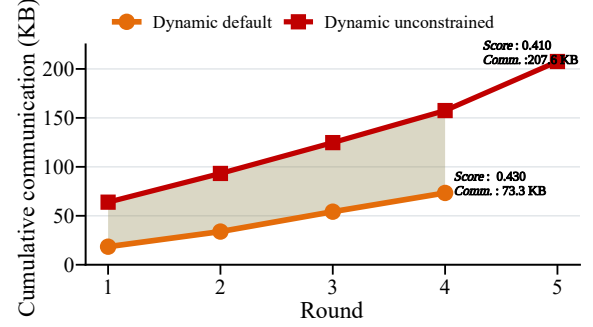


Fig. 12. Cumulative communication under dynamic default and dynamic unconstrained policies on 3GPP-TSG.

objective gaps. In contrast, Utility-only and Channel-only scheduling produce larger gaps by considering only memory utility or wireless cost. These results demonstrate that jointly accounting for memory utility and wireless conditions enables FedMemOpt to achieve near-optimal scheduling decisions.

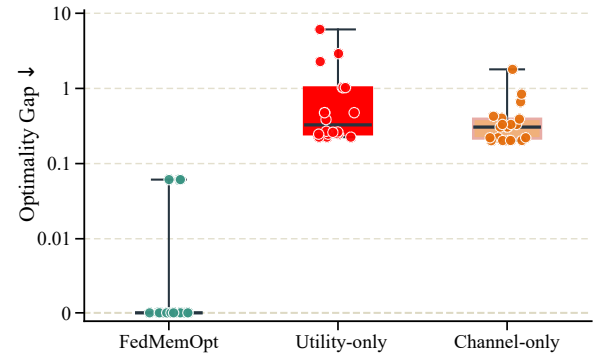


Fig. 13. Objective gaps to exhaustive enumeration over 20 frozen wireless cells. FedMemOpt jointly accounts for memory utility and wireless cost, whereas Value-only and Channel-only isolate one factor.

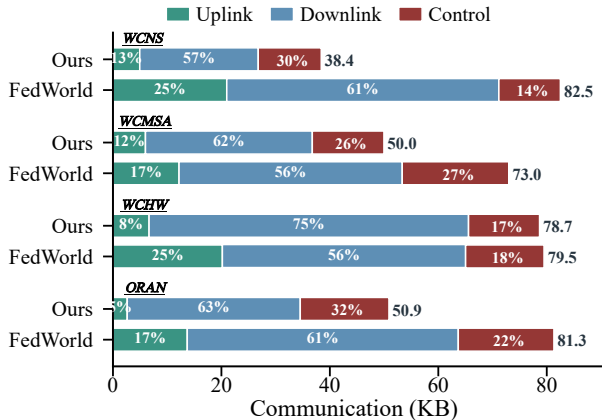


Fig. 11. Communication-cost composition of FedMemOpt and FedWorld. Bar lengths report measured KB; segment labels report component proportions.

E. Case Studies: Distributed Underground Methane-Risk Identification as a Federated Agentic Collaboration Task

To evaluate the practical applicability of FedMemOpt in real-world industrial scenarios, we deploy it in an underground internet of coal mine things (I_oMT) monitoring simulation system for **Distributed Methane-Risk Identification**¹ and contribute the **CoalMine Methane-Risk Identification Dataset**² constructed from real underground-mine multisensor data³. The case study organizes temporally distributed environmental monitoring, ventilation-state, and mining-equipment

¹IoT Simulation: (IOT-SIMULATION-URL)

²CoalMine Dataset: (COALMINE-DATASET-URL)

³Methane—Mendeley: <https://data.mendeley.com/datasets/yd7vw4c5mk/1>

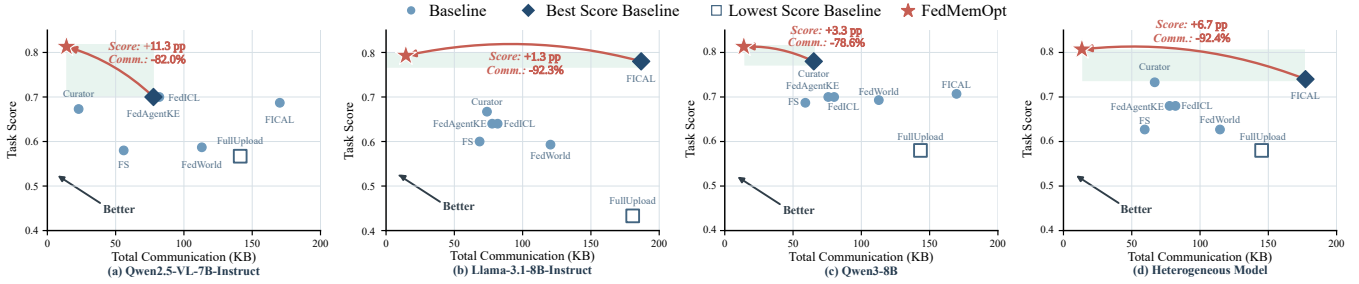


Fig. 14. Performance-communication robustness of FedMemOpt under different edge-node model settings in I_oMT system. FedMemOpt is compared with representative baselines and the best-score baseline in each setting, with the upper-left direction indicating higher task score and lower communication cost.

observations across multiple distributed edge clients, forming a *Federated Agentic Collaboration Task*⁴.

1) *IoMT Simulation Setting*: We construct the simulation from three complementary aspects. **For task formulation**, continuous underground-mine multisensor streams define methane-risk identification as a future-event prediction task. **For federated organization**, temporally separated observations are assigned to multiple edge clients for local agentic-memory construction and collaboration. **For wireless communication**, clients share constrained uplink resources with heterogeneous link efficiencies, while FedMemOpt jointly optimizes memory exchange and resource allocation. Further configuration details are summarized in Table VIII.

TABLE VIII
FEDERATED AGENTIC METHANE-RISK IDENTIFICATION SIMULATION CONFIGURATIONS.

Setting	Configuration
System Deployment	
Central Host	CPU-based orchestration; 2× Intel Xeon Platinum 8358P @ 2.60 GHz, 64 physical cores / 128 threads
Edge-node Platform	GPU-based local inference; 2× NVIDIA RTX A6000, 48 GB/GPU
Default Edge-node Model	Qwen2.5-VL-7B-Instruct, frozen text-only inference
Heterogeneous Deployment	2× Qwen2.5-VL-7B + 2× Qwen3-8B + 1× Llama-3.1-8B-Instruct
Communication Configuration	
Shared uplink bandwidth	500 Hz
Network model	Reliable transmission with zero packet loss
Communication accounting	Uplink, recipient-charged downlink, control, and total bytes
Data Configuration	
Source stream	9,199,930 one-second underground-mine multisensor records
Observation / horizon / lead	60 s / 180 s / 180 s
Risk criterion	Methane concentration $\geq 1.0\%$ across monitored sensors
Data split	60 construction / 40 validation / 150 test

Notes: More detailed simulation configurations are provided in the *IoT Simulation Configurations*⁵.

2) Main Results:

Table IX evaluates the performance of FedMemOpt across different settings. FedMemOpt achieves the best task performance and the lowest positive communication in all settings, improving task performance by 16.1%, 1.7%, and 9.1% over the strongest baselines under Qwen2.5-VL, Llama-3.1, and heterogeneous deployments, respectively. It also reduces communication by 38.9%, 78.8%, and 77.3% relative to the lowest-communication positive baselines. These results validate its effectiveness across practical I_oMT deployment settings.

3) *Adaptability in IoMT*: We further investigate the adaptability of FedMemOpt in I_oMT, focusing on its robustness across different edge-node model deployments and the performance-communication efficiency of agentic-memory evolution.

Edge-node Model Robustness: As shown in Fig. 14, FedMemOpt maintains consistently strong task performance

⁴The Complete Simulation System and Adaptation Interfaces are available at: [PROJECT-URL](#)

⁵IoT Simulation Configurations: [IOT-SIMULATION-URL](#)

TABLE IX
METHANE-RISK IDENTIFICATION UNDER DIFFERENT EDGE-NODE MODEL DEPLOYMENTS IN COALMINE IOT SYSTEM.

Method	Qwen2.5-VL-7B		Llama-3.1-8B		Heterogeneous	
	Score ↑	Comm. ↓	Score ↑	Comm. ↓	Score ↑	Comm. ↓
FS-Compatible [3]	0.580	55.9	0.600	68.5	0.627	59.5
Curator [13]	0.673	22.9	0.667	73.9	0.733	66.8
FedICL [24]	0.700	82.0	0.640	81.7	0.680	82.0
FedWorld [5]	0.587	113.1	0.593	120.4	0.627	114.6
FedAgentKE [4]	0.700	77.7	0.640	77.4	0.680	77.7
FICAL [6]	0.687	170.2	0.780	186.8	0.740	177.2
NoMemory	0.660	—	0.453	—	0.580	—
Local	0.580	—	0.620	—	0.667	—
FullUpload	0.567	141.2	0.433	180.6	0.580	145.1
FedMemOpt	0.813	14.0	0.793	14.5	0.807	13.5

Notes: Comm. denotes total communication in KB. *Heterogeneous* denotes an edge-node deployment with 2×Qwen2.5-VL, 2×Qwen3-8B, and 1×Llama-3.1-8B model. Best results are bold and second-best results are underlined. and stable communication cost across multiple edge-node model settings. Specifically, task-performance variation remains within 2.5%, while communication-cost variation is marginal, demonstrating the adaptability of FedMemOpt to different edge-node model deployments.

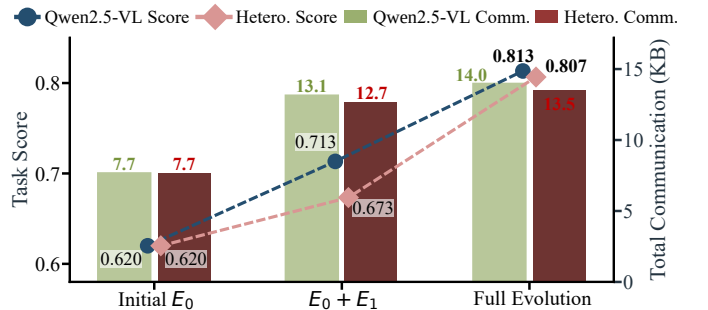


Fig. 15. Task-performance and communication evolution during agentic-memory evolution under different edge-node model settings in the I_oMT system. E_0 denotes the initial experience, $E_0 + E_1$ the memory after first-round evolution, and Full Evolution the complete memory-evolution stage.

Memory Evolution and Communication Efficiency: As shown in Fig. 15, agentic-memory evolution progressively improves task performance across different edge-node model settings with limited communication growth. The final evolution stage further improves task performance by 19.9%, while increasing communication by only 6.3%, demonstrating the communication efficiency of agentic-memory evolution.

Overall, the methane-risk prediction case study based on I_oMT shows that FedMemOpt supports efficient agentic-memory evolution across distributed edge nodes with limited communication growth, enabling effective agent-driven methane-risk prediction and validating the practical adaptability of the communication-optimized federated agentic collaboration in industrial IoT scenarios.

VI. CONCLUSION AND FUTURE WORK

This paper studied *communication-efficient federated agentic-memory evolution framework* over wireless networks and introduced **FedMEvo**, which *jointly optimizes agentic-memory evolution and wireless communication cost* through memory exchange over shared wireless resources. FedMEvo also employs multi-granularity semantic representations of memories to support this joint optimization. We provide a rigorous theoretical characterization of the resulting co-optimization problem together with complete solution analysis and proofs. Extensive experiments demonstrate superior performance–communication tradeoffs, robustness under heterogeneous conditions, and practical applicability in an industrial IoT scenario.

The proposed FedMEvo primarily focuses on round-wise communication-efficient evolution of memory states; *a natural extension is to support a long-horizon memory lifecycle management*. Such long-horizon management offers a natural pathway toward *recursive self-improvement* (RSI) [32]. Rather than evolving only the local memory state $\mathcal{L}_i^t$, future systems may jointly evolve the memory state and an agent self-improvement policy Π_i^t , i.e., $(\mathcal{L}_i^t, \Pi_i^t) \rightarrow (\mathcal{L}_i^{t+1}, \Pi_i^{t+1})$. Here, Π_i^t characterizes **the mechanism that transforms accumulated experience into subsequent capability improvement**. *Such recursive evolution could enable federated agents to continuously refine their adaptation to task dynamics and wireless-network variations*, thereby promoting the co-evolution of task capability and communication adaptability.

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